

How is the story of partitions linked to booting of OS? Well, almost all earlier operating systems (such as Linux 1.2) needed a primary partition to install at least its 'boot' sector. When people used the early Windows OS (such as Windows 95/98), they had to use the FDISK tool to create partitions. This disk partitioning tool by Microsoft did not allow more than one primary partition. Consequently, installing more than one operating system became difficult as installing Linux first and creating a primary partition for Windows installation will not work because Windows will overwrite the MBR. This resulted in the creation of Boot Loader. When the Boot Loader was being formulated, some changes were also made to the partition-system of Unix-like OSes. Eventually, Linux's */boot* or for that matter all partitions can be a logical drive and it will boot!

This is true if it is made to dual boot with another OS. But if it is the only OS then the */boot* partition has to be primary with the MBR in it of course (first cylinder of the hard disk).

Linux Loader or the popular LILO thus became a valuable asset of the Linux community. But later on Grub gained importance as it is more secure and has some features in addition to LILO. The newer distribution does not bundle LILO due to the popularity of Grub.

Assume that an OS is installed, say Windows, prior to Linux. The role of a boot loader is to append the MBR and not overwrite it. It writes its presence to the MBR. When a PC is turned on, the control of booting an OS is transferred by the MBR to the boot loader. The boot loader is set to boot a specific OS within a set time. The default OS can either be Windows or Linux. If Windows is selected to boot, then the boot loader will transfer control back to the MBR to follow up with booting Windows (boot information of Windows is still in the MBR). And, if Linux is chosen, then *init* takes the control from the boot loader. '*init*' is a daemon (daemon is a process, explained later in the chapter) that starts various other system processes required for booting and eventually running the Linux OS.